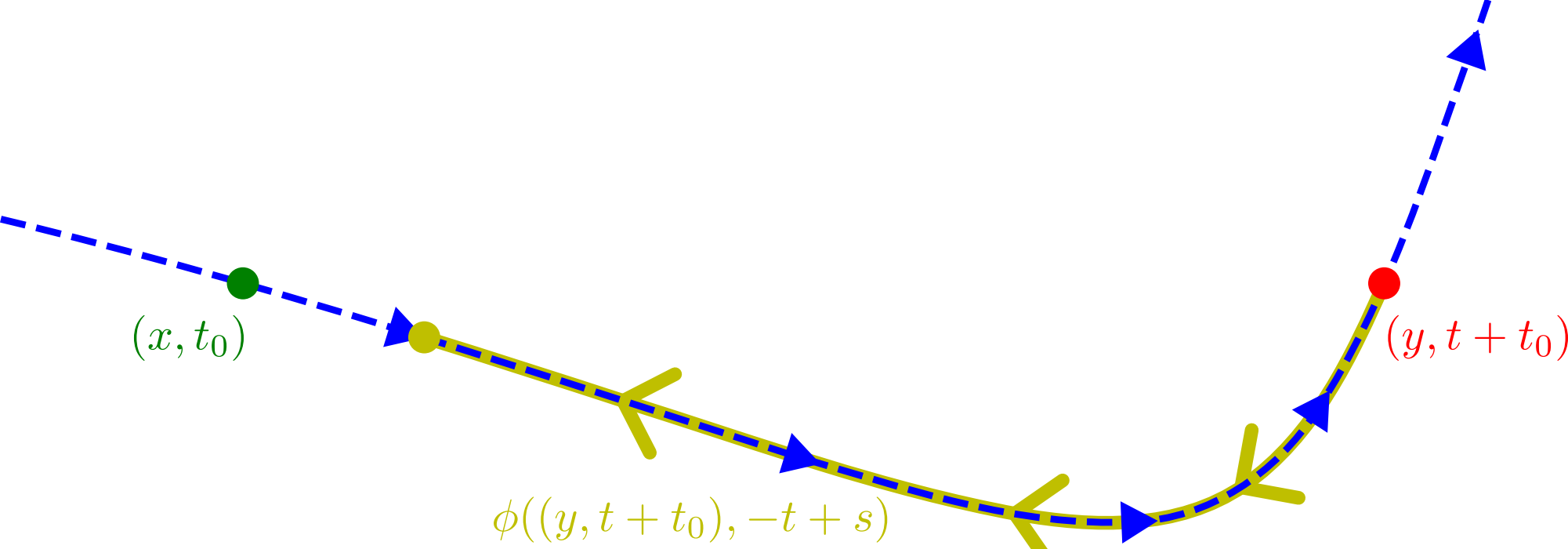
(x, t_0)
$(y, t + t_0)$
$\phi((y, t + t_0), -t + s)$